ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Cornerstone project work entitled *"**Online Shopping Cart Simulation**"* is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Computer Science and Engineering for the academic year 2025-2026.

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DECLARATION

We hereby declare that the Cornerstone project entitled "**ONLINE SHOPPING CART SIMULATION**" is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree** in **Computer Science and Engineering**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

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ABSTRACT

The Online Shopping Cart Simulation is a desktop-based application designed to efficiently manage the process of selecting, adding, and purchasing products in an online shopping environment by replacing traditional manual purchasing methods with a digital solution. The system enables users to browse products, add items to the cart, update quantities, remove items, and proceed to checkout in a structured and secure manner using a relational database. Developed using Core Java, Swing for the graphical user interface, JDBC for database connectivity, and MySQL Database for backend storage, the application ensures data accuracy, minimizes redundancy, and improves user experience. With its user-friendly interface and efficient transaction handling capabilities, the system enhances shopping convenience and provides a reliable solution for simulating an online purchasing system.

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ACRONYMS

- **OSCS** : Online Shopping Cart Simulation
- **JDBC** : Java Database Connectivity
- **GUI** : Graphical User Interface
- **SQL** : Structured Query Language
- **DBMS** : Database Management System
- **UI** : User Interface

1. INTRODUCTION

1.1 Brief Information about the Project

The Online Shopping Cart Simulation (OSCS) is a desktop-based application developed to simulate an e-commerce platform where users can browse products, add items to a cart, and place orders efficiently. It provides a centralized system to manage users, products, cart operations, and orders in a structured manner. The system replaces traditional manual purchasing methods with a digital solution, thereby improving accuracy, reducing redundancy, and enhancing user experience.

The application is developed using Core Java for programming, Swing for designing the graphical user interface, JDBC for database connectivity, and MySQL database for storing data such as user details, products, and orders. It is designed to be user-friendly and suitable for learning and implementing real-world online shopping concepts.

1.2 Motivation and Contribution of Project

In many traditional shopping systems, users face limitations such as lack of automation, difficulty in managing product availability, and inefficient tracking of orders. Manual processes can lead to errors, data inconsistency, and poor user experience. These challenges motivated the development of the Online Shopping Cart Simulation.

The key contributions of this project include:

- Providing a digital platform for online product browsing and purchasing
- Reducing manual effort in managing shopping and order processes
- Improving accuracy and consistency of product and transaction data
- Enabling quick search, selection, and ordering of products
- Supporting multiple roles such as Admin, User, and Supplier
- Enhancing data security through role-based access control

This system simplifies online shopping operations and improves overall efficiency.

1.3 Objectives of the Project

The main objectives of the Online Shopping Cart Simulation are:

- To develop a user-friendly application for online shopping

- To store product, user, and order information in a structured database
- To provide functionalities for adding products to cart and placing orders
- To enable users to browse and search products easily
- To manage inventory and product details efficiently
- To display order and cart information in an organized format

1.4 Scope of the Project

The Online Shopping Cart Simulation can be used as a learning tool for understanding e-commerce systems and database-driven applications. It supports operations such as user registration and login, product browsing, adding items to the cart, placing orders, and managing inventory through supplier and admin roles.

The system provides a simple and user-friendly interface for handling online shopping activities in a centralized database. It is suitable for educational purposes and can be extended into a full-scale e-commerce application with additional features such as payment integration, delivery tracking, and recommendation systems.

2. SYSTEM ANALYSIS

2.1 Functional Requirements

Functional requirements define the core operations and functionalities that the Online Shopping Cart Simulation (OSCS) must perform to efficiently manage online shopping activities.

The functional requirements of OSCS are:

User Registration and Login

The system should allow users to create new accounts and securely log in using their credentials. It should also support password recovery using security questions.

Product Management

The system should allow the admin and suppliers to add, update, and delete product details such as product name, description, price, stock, and images.

Browse and Search Products

The system should enable users to browse available products and search for items using keywords or categories.

Shopping Cart Management

The system should allow users to add products to the cart, update quantities, and remove items before placing an order.

Order Placement

The system should enable users to place orders for selected cart items. It should calculate the total amount and store order details.

Wallet Management

The system should maintain a wallet balance for each user and deduct the amount during order placement.

Order Management

The system should allow users to view their order history and track order details such as status and date.

Admin Dashboard

The admin should be able to manage users, products, suppliers, and orders through a centralized dashboard.

Supplier Management

The system should allow suppliers to manage their product inventory, including adding, updating, and deleting products.

2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards of the system.

Performance

The system should respond quickly to user actions such as browsing products, adding items to the cart, and placing orders.

Security

The system should ensure secure authentication and role-based access control for users, admins, and suppliers.

Usability

The system should provide a simple and user-friendly graphical interface for easy navigation and operation.

Reliability

The system should function consistently without crashes and ensure that no data is lost during operations.

Scalability

The system should handle an increasing number of users, products, and orders without performance degradation.

2.3 Requirements Specification

This section specifies the hardware and software required to run the system.

2.3.1 Minimum Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB (minimum)
- Hard Disk: 20 GB free space
- Monitor: Standard display
- Keyboard and Mouse

2.3.2 Software Requirements

- Operating System: Windows 7/10/11
- Programming Language: Java
- IDE: VS Code / IntelliJ IDEA
- Database: MySQL
- Connectivity: JDBC

3. TECHNOLOGY DESCRIPTION

3.1 Programming Language

3.1.1 Introduction to Java

Java is a high-level, object-oriented programming language developed by Sun Microsystems. It is widely used for developing desktop, web, and enterprise applications due to its platform independence, security, and robustness.

In the Online Shopping Cart Simulation, Java is used to implement the core functionality of the application. It handles operations such as user authentication, product management, cart handling, and order processing. Java provides features like object-oriented programming, exception handling, and strong memory management, which help in building reliable and efficient applications.

One of the main advantages of Java is its platform independence, meaning the application can run on any system with a Java Virtual Machine (JVM). This makes it suitable for developing portable applications like the Online Shopping Cart Simulation.

3.2 Framework / Libraries

3.2.1 Introduction to Swing

Swing is a part of Java used to create graphical user interfaces (GUI). It provides a rich set of components such as buttons, labels, text fields, tables, and frames to design user-friendly applications.

In this project, Swing is used to design the user interface of the Online Shopping Cart Simulation. It allows users to interact with the system easily through various windows such as:

- Login Window
- Registration (Sign Up) Window
- Product Browsing Screen
- Shopping Cart Screen
- Order Management Screen
- Admin Dashboard

- Supplier Dashboard

Swing provides flexibility in designing layouts and improves the overall user experience. It makes the application visually interactive and easy to use.

3.3 Database Technology

The Online Shopping Cart Simulation uses a relational database to store and manage data such as users, products, cart items, and orders in a structured manner. The database ensures efficient data storage, retrieval, and manipulation, which is essential for maintaining accuracy and consistency.

In this project, MySQL Database is used as the backend, and JDBC is used for connecting the Java application to the database.

3.3.1 JDBC (Java Database Connectivity)

JDBC is an API (Application Programming Interface) used in Java to connect and interact with databases. It allows Java applications to execute SQL queries and retrieve or update data.

In the Online Shopping Cart Simulation, JDBC is used to connect the Java application with the MySQL database. It performs operations such as:

- Establishing a connection to the database
- Executing SQL queries (INSERT, UPDATE, DELETE, SELECT)
- Retrieving data from the database
- Closing the database connection

JDBC acts as a bridge between the application and the database, ensuring smooth communication and efficient data handling.

3.3.2 MySQL Database

MySQL is a widely used relational database management system that stores and manages structured data efficiently. It provides high performance, reliability, and security.

In this project, MySQL is used to store data such as user details, product information, cart items, and order records. It supports SQL for data manipulation and ensures data integrity through constraints like primary keys, foreign keys, and data types.

3.4 Tools & IDEs

3.4.1 Apache NetBeans

Apache NetBeans is an integrated development environment (IDE) used for developing Java applications. It provides a powerful platform for building desktop, web, and enterprise applications with ease.

In the Online Shopping Cart Simulation, Apache NetBeans is used as the development tool to write, compile, and run the Java code. It offers features such as drag-and-drop GUI design, syntax highlighting, code completion, and debugging support, which help in improving development efficiency.

NetBeans also simplifies the creation of Swing-based user interfaces through its built-in GUI builder, making it easier to design forms like login screens, dashboards, and product management panels. Overall, Apache NetBeans provides a user-friendly environment that enhances productivity and simplifies application development.

4. DATABASE DESIGN

4.1 Database Tables

The Online Shopping Cart Simulation uses a relational database to store data such as users, products, cart items, and orders in a structured manner. The database consists of multiple tables:

- The **Users** table stores information about users, admins, and suppliers
- The **Wallet** table stores user balance details
- The **Products** table stores product information
- The **Cart** table stores temporary selected items by users
- The **Orders** table stores order details
- The **Order_Items** table stores individual items of each order

This structured design helps in reducing redundancy and improving data organization.

4.2 Table Structure

4.2.1 Users Table

Field Name	Data Type	Description	Constraint
id	INT	User ID	Primary Key
username	VARCHAR(50)	Username	Unique, Not Null
email	VARCHAR(100)	Email Address	Unique, Not Null
password	VARCHAR(255)	Password	Not Null
mobile	VARCHAR(15)	Mobile Number	–
age	INT	User Age	–
address	TEXT	Address	–
security_question	VARCHAR(255)	Security Question	–
security_answer	VARCHAR(255)	Security Answer	–
role	ENUM	User Role (admin/user/supplier)	Default 'user'
approved	TINYINT	Approval Status	Default 1

Table 1: Users Relational Database Table

4.2.2 Wallet Table

Field Name	Data Type	Description	Constraint
user_id	INT	User ID	Primary Key, Foreign Key
balance	DOUBLE	Wallet Balance	Default Value

Table 2: Wallet Relational Database Table

4.2.3 Products Table

Field Name	Data Type	Description	Constraint
id	INT	Product ID	Primary Key
name	VARCHAR(255)	Product Name	–
description	TEXT	Product Description	–
price	DOUBLE	Product Price	–
stock	INT	Product Stock	–
image_path	VARCHAR(255)	Image Path	–
supplier_id	INT	Supplier ID	Foreign Key

Table 3: Products Relational Database Table

4.2.4 Cart Table

Field Name	Data Type	Description	Constraint
id	INT	Cart ID	Primary Key
user_id	INT	User ID	Foreign Key
product_id	INT	Product ID	Foreign Key
quantity	INT	Quantity	Default 1
price	DOUBLE	Product Price	–

Table 4: Cart Relational Database Table

4.2.5 Orders Table

Field Name	Data Type	Description	Constraint
id	INT	Order ID	Primary Key
user_id	INT	User ID	Foreign Key
total_amount	DOUBLE	Total Amount	–
status	VARCHAR(40)	Order Status	Default 'Confirmed'
order_date	TIMESTAMP	Order Date	Default Current Time

Table 5: Orders Relational Database Table

4.2.6 Order Items Table

Field Name	Data Type	Description	Constraint
id	INT	Order Item ID	Primary Key
order_id	INT	Order ID	Foreign Key
product_id	INT	Product ID	Foreign Key
quantity	INT	Quantity	–
price	DOUBLE	Product Price	–

Table 6: Order Items Relational Database Table

4.3 Keys and Constraints

Keys and constraints are used to maintain data integrity and ensure accurate data storage.

Primary Keys

- id in Users, Products, Cart, Orders, and Order_Items tables
- user_id in Wallet table

Foreign Keys

- user_id in Wallet, Cart, and Orders tables references Users table
- product_id in Cart and Order_Items references Products table
- order_id in Order_Items references Orders table
- supplier_id in Products references Users table

NOT NULL Constraints

- Important fields such as username, email, and password cannot be empty

UNIQUE Constraints

- Username and Email must be unique

Default Constraints

- Wallet balance and order status have default values

These constraints ensure data consistency and prevent invalid or duplicate entries.

4.4 Table Relationships

The tables in the system are interconnected to represent real-world relationships:

- One user can have one wallet (One-to-One relationship)
- One user can have multiple cart items (One-to-Many relationship)

- One user can place multiple orders (One-to-Many relationship)
- One order can contain multiple products (One-to-Many relationship via Order_Items)
- One supplier can manage multiple products (One-to-Many relationship)

These relationships help in organizing data efficiently and ensure smooth operation of the system.

5. SYSTEM DESIGN

5.1 System Architecture

The Online Shopping Cart Simulation follows a three-layer architecture:

1. Presentation Layer (User Interface)

- Developed using Java Swing
- Allows users (Admin, User, Supplier) to interact with the system
- Provides screens such as Login, Sign Up, Product Browsing, Cart, and Dashboards

2. Application Layer (Business Logic)

- Processes user inputs and actions
- Handles operations such as product management, cart operations, order placement, and user authentication
- Implements business rules like stock validation, wallet deduction, and role-based access

3. Data Layer (Database)

- Stores data related to users, products, cart, and orders
- Managed using MySQL database
- Accessed through JDBC for executing SQL queries

This architecture ensures better organization, security, scalability, and maintainability of the system.

5.2 Modules Description

The system is divided into the following modules:

1. User Management Module

- Handles user registration and login
- Supports password recovery using security questions
- Manages user roles (Admin, User, Supplier)

2. Product Management Module

- Allows admin and suppliers to add new products
- Updates product details such as price, stock, and description

- Deletes products when necessary
- Displays products in a structured format

3. Shopping Cart Module

- Allows users to add products to the cart
- Updates product quantity
- Removes items from the cart
- Calculates total cost of selected items

4. Order Management Module

- Handles order placement process
- Stores order details and order items
- Updates product stock after order
- Displays order history and status

5. Wallet Module

- Maintains user wallet balance
- Deducts amount during order placement
- Ensures sufficient balance before purchase

6. Admin Module

- Manages users, suppliers, products, and orders
- Approves supplier accounts
- Monitors overall system activity

7. Supplier Module

- Allows suppliers to manage their product inventory
- Adds, updates, and deletes products
- Views orders related to their products

8. Search Module

- Enables users to search products using keywords or categories
- Provides quick and accurate results

9. Database Connectivity Module

- Connects Java application with MySQL database using JDBC
- Executes SQL queries (INSERT, UPDATE, DELETE, SELECT)

- Ensures smooth data communication

10. User Interface Module

- Provides GUI using Java Swing
- Handles user interactions through forms and dashboards
- Ensures a user-friendly experience

6. SYSTEM IMPLEMENTATION

Login Screen

Displays the login interface where the user enters username and password to access the system. It also provides options for registration and password recovery.

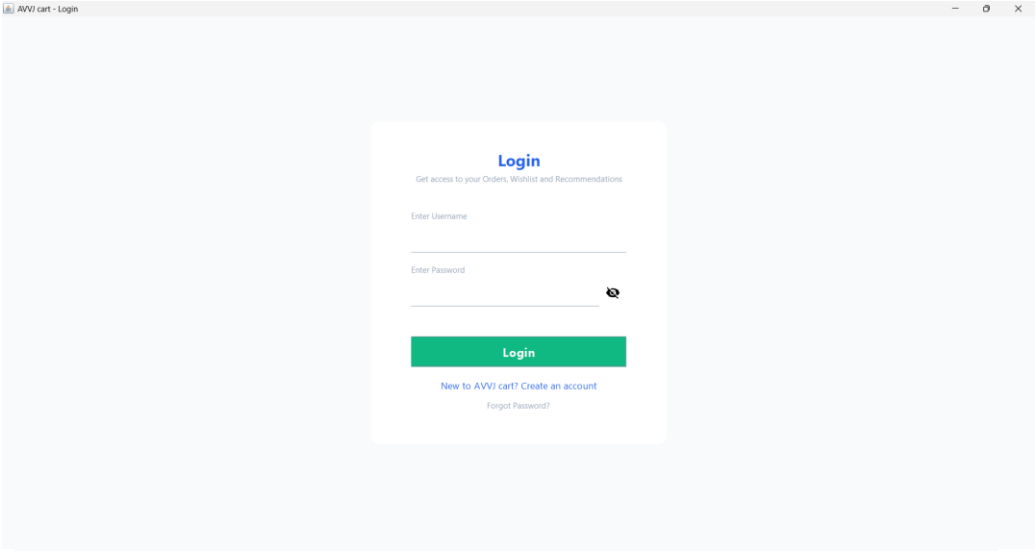

Figure 1: Login Screen of Online Shopping Cart Simulation

Registration (Sign Up) Screen

Provides a form for new users to create an account by entering details such as username, email, password, mobile number, and security question.

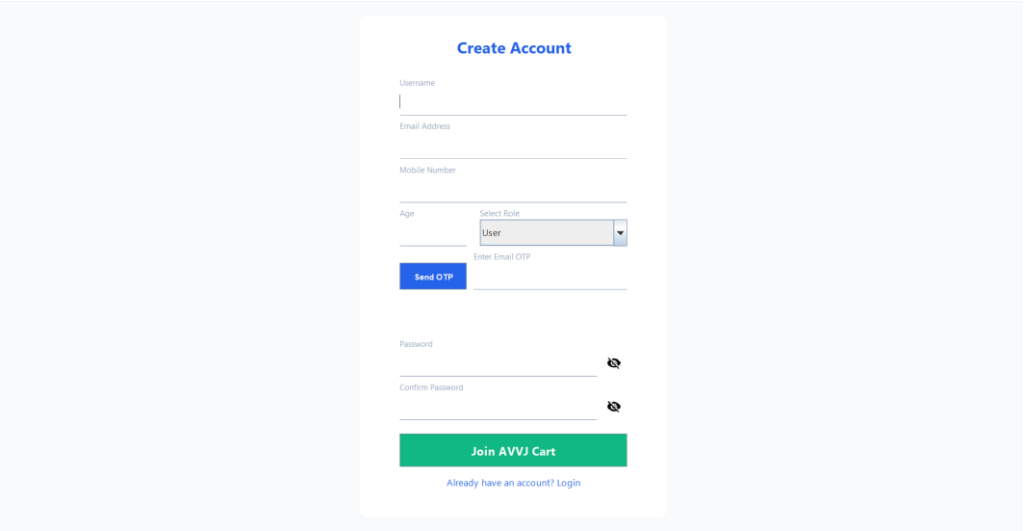

Figure 2: Registration (Sign Up) Screen of Online Shopping Cart Simulation

Forgot Password Screen

Allows users to reset their password by verifying their identity using a security question and answer.

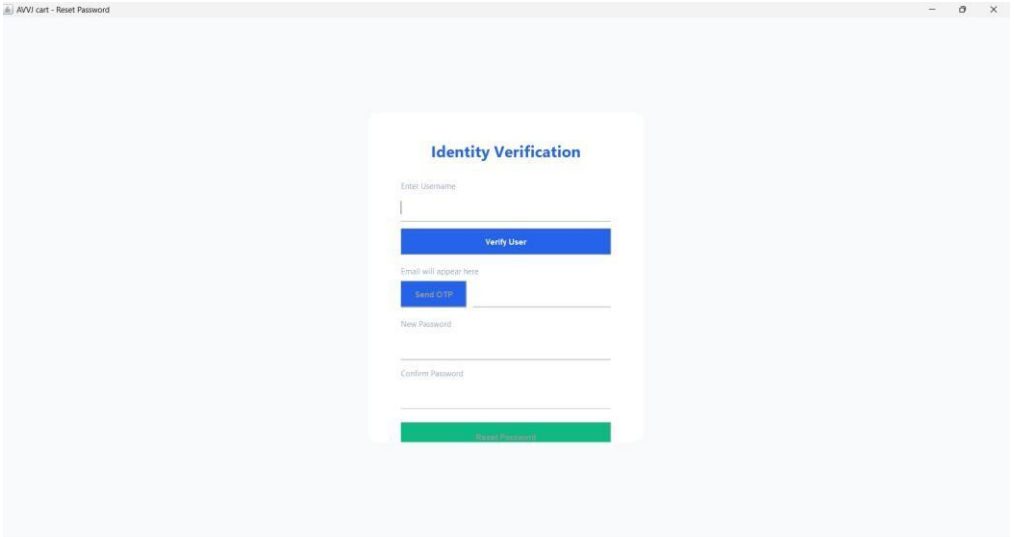

Figure 3: Forgot Password Screen of Online Shopping Cart Simulation

Admin Dashboard Screen

Displays administrative controls such as managing users, products, orders, and suppliers. It provides a centralized view of system activities.

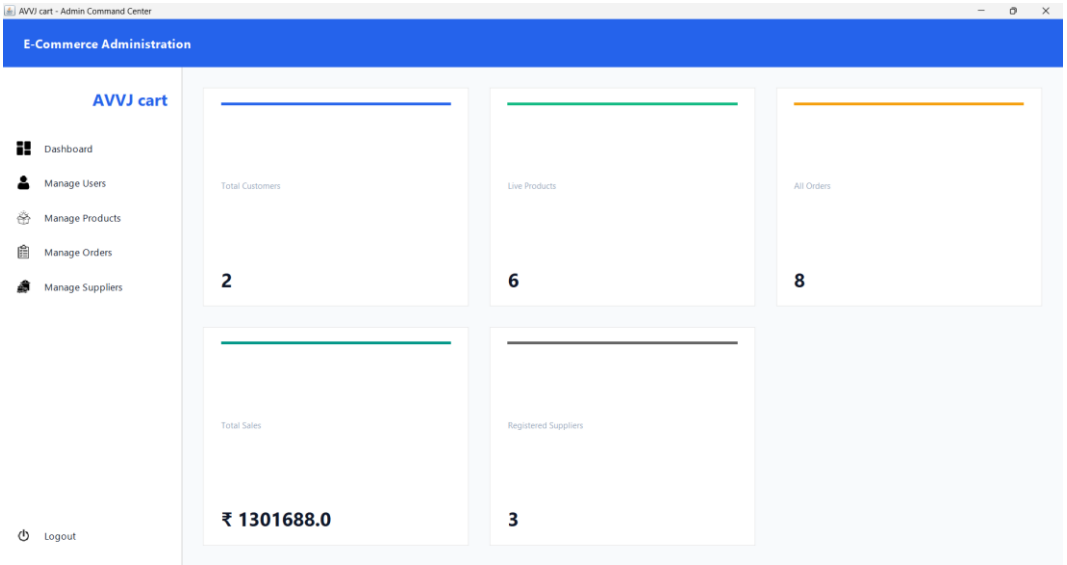

Figure 4: Admin Dashboard Screen of Online Shopping Cart Simulation

User Dashboard Screen

Displays user-specific information such as wallet balance, cart summary, orders, and navigation options for shopping.

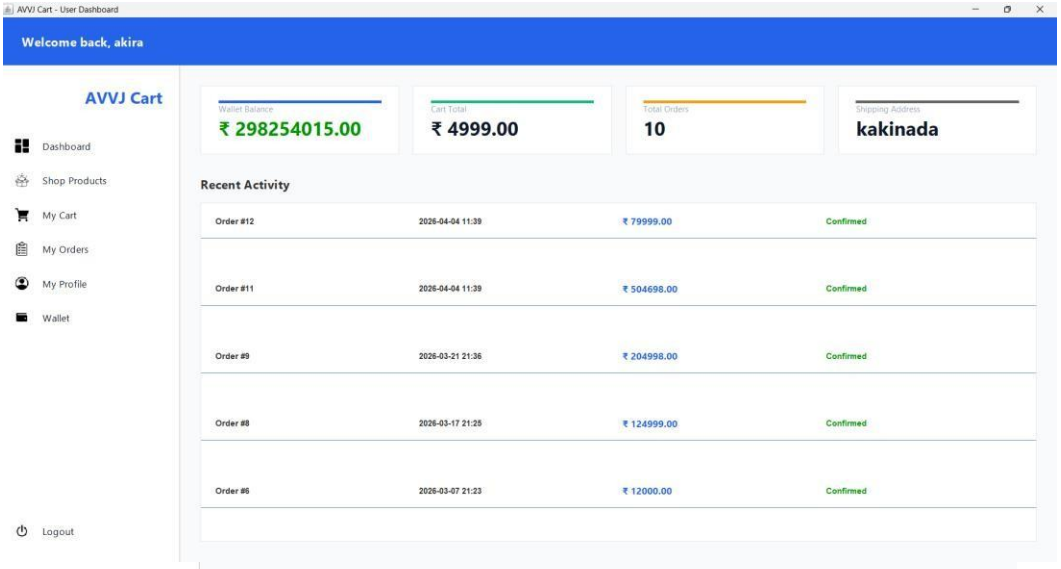

Figure 5: User Dashboard Screen of Online Shopping Cart Simulation

Product Browsing Screen

Allows users to view available products, search items, and filter by categories. Users can add products to the cart from this screen.

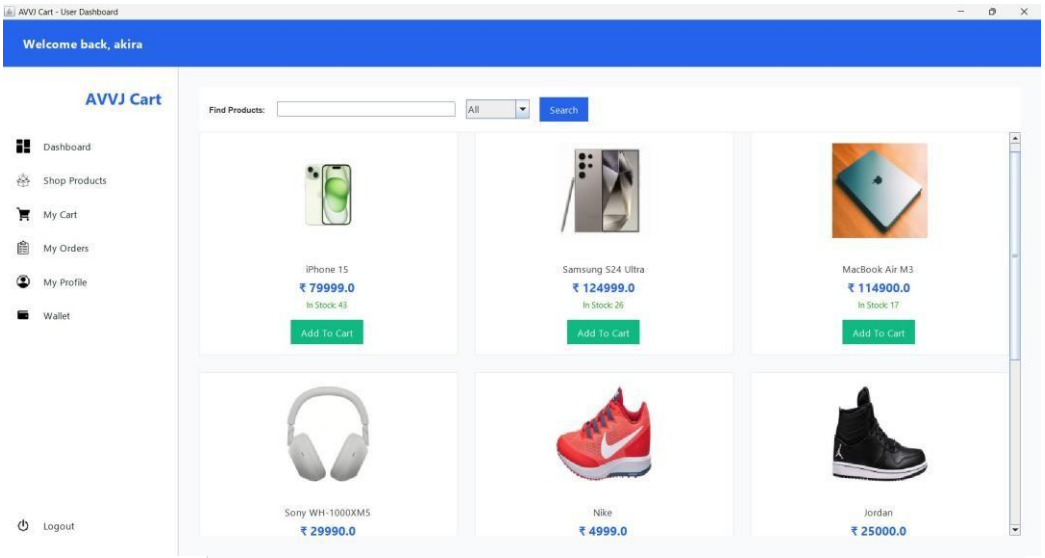

Figure 6: Product Browsing Screen of Online Shopping Cart Simulation

Shopping Cart Screen

Displays selected products along with quantity, price, and total amount. Users can update quantity, remove items, and proceed to checkout.

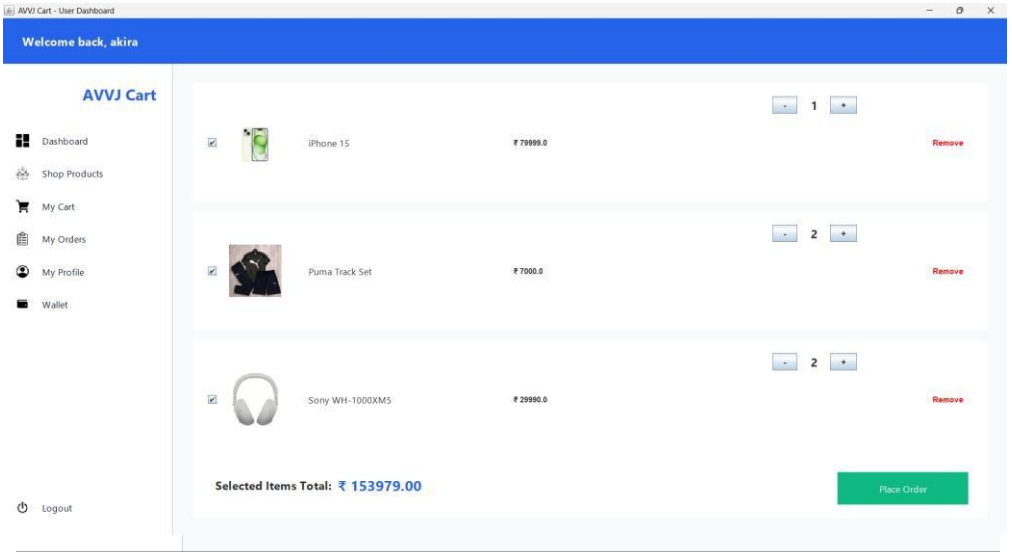

Figure 7: Shopping Cart Screen of Online Shopping Cart Simulation

Order Placement Screen

Handles the process of placing orders, deducting wallet balance, and confirming purchases.

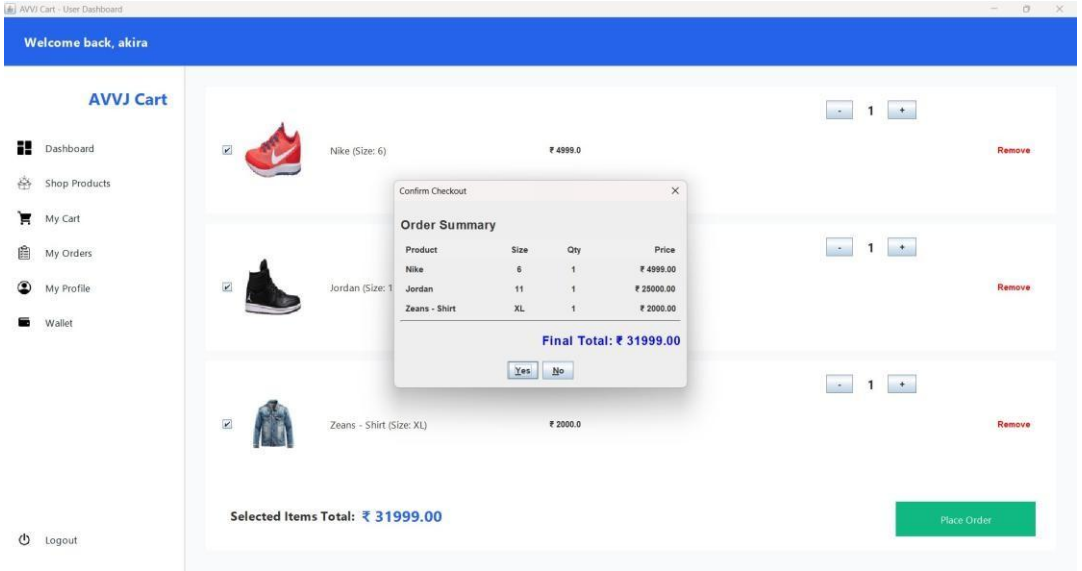

Figure 8: Order Placement Screen of Online Shopping Cart Simulation

Supplier Dashboard Screen

Allows suppliers to manage their products, view orders, and monitor inventory.

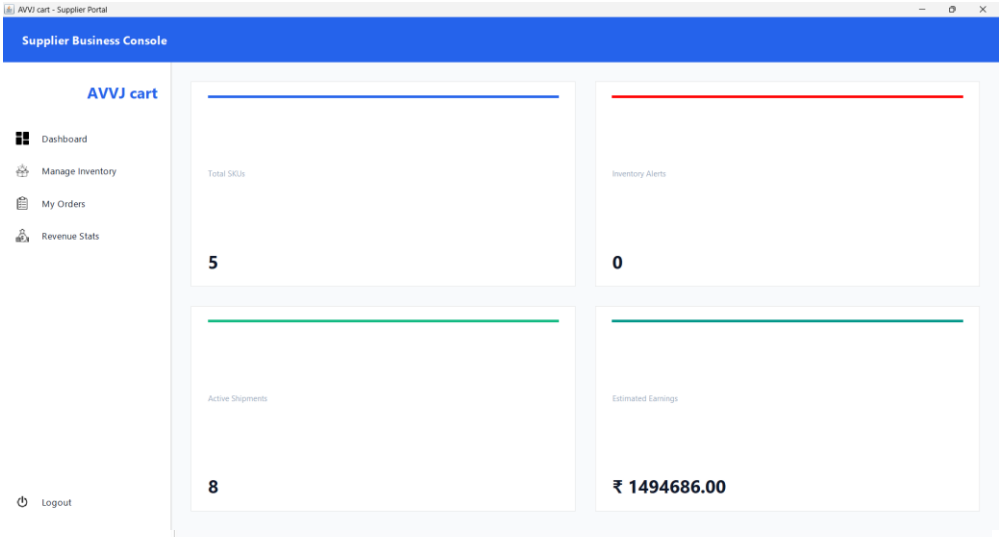

Figure 9: Supplier Dashboard Screen of Online Shopping Cart Simulation

Inventory Management Screen

Displays product inventory for suppliers, enabling them to add, update, or delete products.

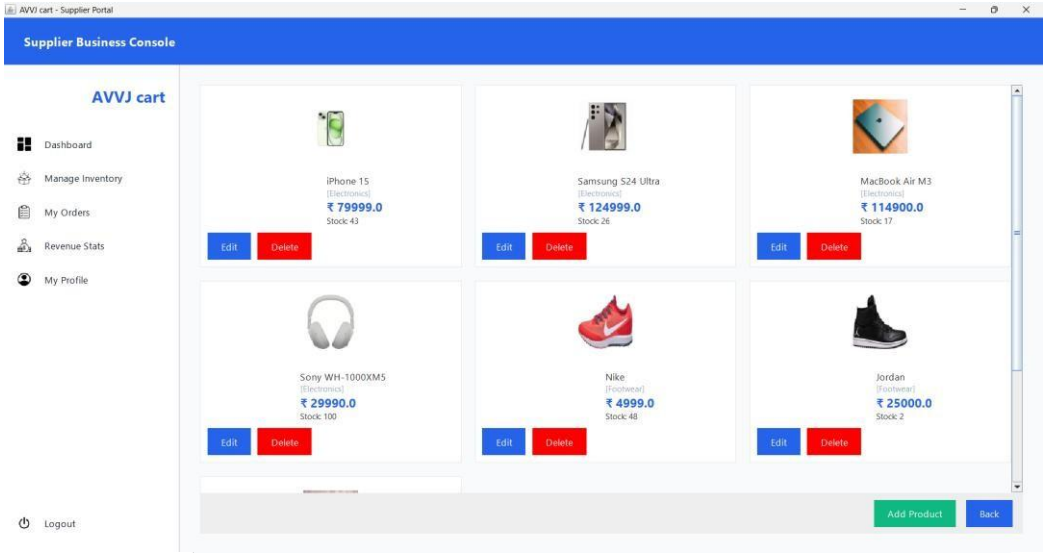

Figure 10: Inventory Management Screen of Online Shopping Cart Simulation

7. CONCLUSION

The Online Shopping Cart Simulation has been successfully developed to simulate an efficient and organized online shopping system. The application replaces traditional manual shopping processes with a digital solution, thereby improving accuracy, saving time, and enhancing user convenience.

The system provides essential functionalities such as user registration and login, product browsing, cart management, and order placement through a user-friendly interface. By using technologies like Java, Swing, JDBC, and MySQL Database, the application ensures reliable performance, secure data handling, and efficient transaction management.

The project also supports multiple roles such as Admin, User, and Supplier, which enhances the system's flexibility and real-world applicability. Features like wallet management, inventory control, and order tracking further improve the overall functionality of the system.

Overall, the system simplifies online shopping operations and reduces manual effort while ensuring data consistency and better user experience. It also provides a scalable platform that can be enhanced in the future by adding features such as online payment integration, delivery tracking, recommendation systems, and web-based access.

In conclusion, the Online Shopping Cart Simulation meets the project objectives and serves as an effective tool for understanding and implementing real-world e-commerce systems.

8. FUTURE ENHANCEMENTS

The Online Shopping Cart Simulation developed in this project provides essential functionalities for simulating an e-commerce system. However, the system can be further enhanced by adding advanced features to improve its functionality and user experience.

Some possible future enhancements include:

Online Payment Integration

Integrating payment gateways such as UPI, credit/debit cards, and net banking for real-time transactions.

Delivery Tracking System

Providing order tracking features to allow users to monitor the delivery status of their orders.

Web-Based Application

Converting the desktop application into a web-based or cloud-based system for remote access.

Recommendation System

Implementing AI-based product recommendations based on user preferences and purchase history.

Multi-User Scalability

Enhancing the system to support a large number of concurrent users efficiently.

Improved User Interface

Enhancing the design with modern UI/UX techniques for better user experience.

Notification System

Adding email or SMS notifications for order confirmation and updates.

Advanced Security

Implementing encryption, secure authentication, and role-based access control for better data protection.

These enhancements can make the system more scalable, secure, and suitable for real-world e-commerce applications.

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APPENDIX

A.1 Sample Code

Database Connection Code

The following code establishes a connection between the Java application and the MySQL database using JDBC.

DBConnection.java:

```
import java.sql.*;
public class DBConnection {
    public static Connection getConnection() {
        Connection con = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/avvj_cart",
                "root",
                "password"
            );
        } catch (Exception e) {
            System.out.println(e);
        }
        return con;
    }
}
```

Insert Product Record

This code is used to add a new product into the database.

InsertProduct.java:

```
import java.sql.*;
public class InsertProduct {
    public static void main(String[] args) {
        try {
            Connection con = DBConnection.getConnection();
            String query = "INSERT INTO products(name, description, pric
e, "
                        + "stock, image_path, supplier_id) VALUES(?,?,?
,?,?,?)";
```

```
            PreparedStatement pst = con.prepareStatement(query);
            pst.setString(1, "Smart Watch");
            pst.setString(2, "Fitness tracking smart watch");
            pst.setDouble(3, 2999);
            pst.setInt(4, 50);
            pst.setString(5, "/icons/watch.png");
            pst.setInt(6, 2);
            int i = pst.executeUpdate();
            if (i > 0) System.out.println("Product Added Successfully");
            con.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

A.2 SQL Queries

Create Users Table

```
CREATE TABLE users(
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50) UNIQUE NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    password VARCHAR(255) NOT NULL,
    mobile VARCHAR(15),
    age INT,
    address TEXT,
    security_question VARCHAR(255),
    security_answer VARCHAR(255),
    role ENUM('user','admin','supplier') DEFAULT 'user',
    approved TINYINT(1) DEFAULT 1
);
```

Create Products Table

```
CREATE TABLE products(
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(255),
    description TEXT,
    price DOUBLE,
    stock INT,
    image_path VARCHAR(255),
    supplier_id INT,
    FOREIGN KEY(supplier_id) REFERENCES users(id)
);
```

Create Orders Table

```
CREATE TABLE orders(
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
    total_amount DOUBLE,
    status VARCHAR(40) DEFAULT 'Confirmed',
    order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY(user_id) REFERENCES users(id)
);
```

Create Cart Table

```
CREATE TABLE cart(
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
```

```
    product_id INT,
    quantity INT DEFAULT 1,
    price DOUBLE,
    FOREIGN KEY(user_id) REFERENCES users(id),
    FOREIGN KEY(product_id) REFERENCES products(id)
);
```